

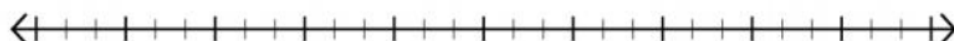
Grade 6 Accelerated
Unit 14
Lesson 10 Homework

Name _____
Date _____
Period _____

- When flying on an airplane, your travel bag can weigh no more than 50 lbs. Your bag currently weighs 27.3 lbs. The inequality $27.3 + m \leq 50$ represents the situation, where m represents the amount of weight that can be added to your bag. What is the maximum amount of weight that can be added to your bag while still making the weight requirement?
- You earn \$8.75 per hour for walking dogs in your neighborhood. You are saving up your dog walking money to purchase a new kayak that costs \$280. The inequality $\$8.75h \geq \280 represents the situation, where h represents the number of hours worked. What is the minimum number of hours you must walk in order to purchase the kayak?
- Complete the table by substituting each value of x to determine if that is a solution to each inequality.

Value of x	What is the value of $-4.5x$	Is $-4.5x > 18$ true or false?	Solution?
-4			☐ Yes ☐ No
-1.5			☐ Yes ☐ No
3			☐ Yes ☐ No
$6\frac{1}{5}$			☐ Yes ☐ No

- Graph the inequality on the number line.



5. Find the solution to both inequalities using the division property of inequality.

$-7y \leq -35$	$7y \leq -35$
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6. Find the solution to both inequalities using the multiplication property of inequality.

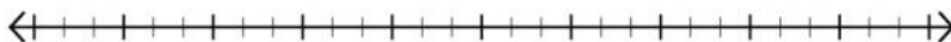
$\frac{y}{3} > 2$	$\frac{y}{-3} > 2$
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Solve each inequality. Then, graph its solution set.

7. $\frac{r}{-4} < 7$



8. $-0.4t \geq -12$



9. $\frac{m}{-2} > \frac{5}{2}$



10. Explain why the direction of the inequality symbol must be reversed when multiplying or dividing by the same negative number.